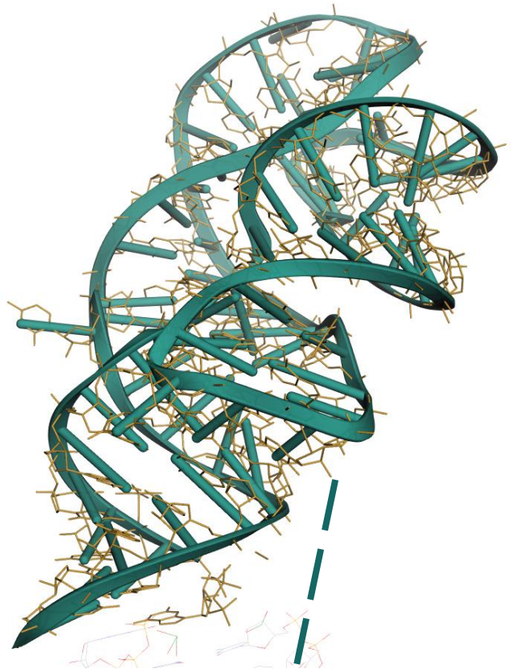
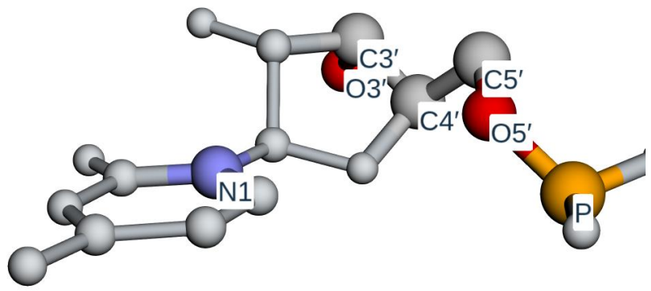
a

RNA tertiary structure

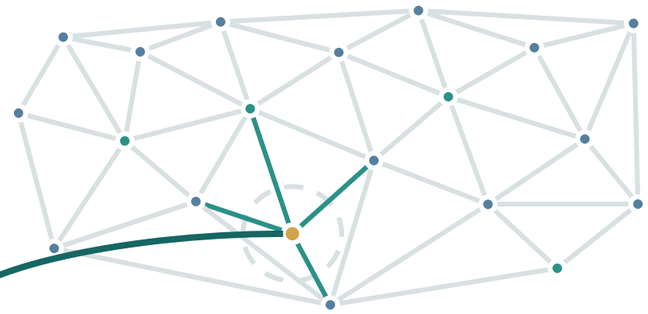


seven-atom model input

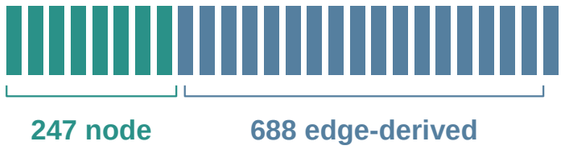



P · O5′ · C5′ · C4′ · C3′ · O3′ · N1/N9

k = 32 nearest-neighbour residue graph



935 geometric descriptors per residue






Task-specific XGBoost

classification / regression

Task-specific readouts

A C G U

**Nucleotide identity**

4-class probabilities · confidence



**Base-pair state**

classification



**Base stacking**

classification



**Secondary-structure class**

3-class classification



**Solvent accessibility**

regression · relative SASA



**Reactivity target**

regression · RibonanzaNet

## b Recovery by sequence population

temporal test




Separate training evaluation: 70.0 family-permissive / 50.4 family-disjoint

## c Confidence-threshold transport